

Mixed Effect Models for Nonnormal Responses

13.1 Generalized Linear Mixed Models

Generalized linear mixed models (GLMM) combine the ideas of generalized linear models with the random effects modeling ideas of the previous two chapters. The response is a random variable, Y_i , taking observed values, y_i , for $i = 1, \dots, n$, and follows an exponential family distribution as defined in Chapter 8:

$$f(y_i|\theta_i, \phi) = \exp \left[\frac{y_i \theta_i - b(\theta_i)}{a(\phi)} + c(y, \phi) \right]$$

Let $EY_i = \mu_i$ and let this be connected to the linear predictor η_i using the link function g by $\eta_i = g(\mu_i)$. Suppose for simplicity that we use the canonical link for g so that we may make the direct connection that $\theta_i = \mu_i$.

Now let the random effects, γ , have distribution $h(\gamma|V)$ for parameters V . The fixed effects are β . Conditional on the random effects, γ ,

$$\theta_i = x_i^T \beta + z_i^T \gamma$$

where x_i and z_i are the corresponding rows from the design matrices, X and Z , for the respective fixed and random effects. Now the likelihood may be written as:

$$L(\beta, \phi, V|y) = \prod_{i=1}^n \int f(y_i|\beta, \phi, \gamma) h(\gamma|V) d\gamma$$

Typically the random effects are assumed normal: $\gamma \sim N(0, D)$. However, unless f is also normal, the integral remains in the likelihood, which becomes difficult to compute, particularly if the random effects structure is complicated.

13.2 Inference

A variety of approaches are available for estimating and performing inference for these models. All have strengths and weaknesses so it is not possible to recommend a single method to use in all circumstances. We present an overview of the theory behind these approaches before demonstrating the implementation on two examples. Later in the chapter, we discuss a related method called generalized estimating equations (GEE).

Penalized Quasi-Likelihood (PQL): In Section 8.2, we described a method by

which GLMs can be fit using only LMs with weights. The idea is to produce a linearized version of the response which we called the adjusted dependent variable (sometimes called the pseudo or working response) defined as

$$\tilde{y}^i = \hat{\eta}^i + (y - \hat{\mu}^i) \frac{d\eta}{d\mu} \Big|_{\hat{\eta}^i}$$

The superscripted *i* indicate the iteration in the optimization algorithm. We have $E(\tilde{y}_i|\gamma) = x_i^T \beta + z_i^T \gamma$ and we may derive an expression for $\text{var}(\tilde{y}_i|\gamma)$. We are now able to use LMM methods with appropriate weighting. Iteration is necessary as $\tilde{y}$ must be updated after each linear mixed model (LMM) fit. This and similar methods are described in Schall (1991) and Breslow and Clayton (1993). The name quasi-likelihood is not entirely appropriate for PQL as we still use the distributional assumptions. The GEE method described later in this chapter fits better with the “quasi” paradigm.

The PQL method has the advantage of relatively easy implementation given that existing LMM methods can be adapted. However, the inference is only asymptotically correct. Biased estimates are mostly likely to arise for binomial responses with small groups (covariate classes) and will be worst for Bernoulli responses. Similar problems will be observed for Poisson response data where the counts tend to be low. Further difficulties will arise with hypothesis testing and confidence intervals because the problems already present in LMMs are added to the approximations introduced by the linearization. We can compute *p*-values using likelihood theory-based methods but we will have limited trust in their veracity. Even so, PQL will tend to be faster and work on more complex models than some of the competitors.

Numerical Integration: Provided the dimension of the random effects γ is not too large, it is possible to use numerical methods to approximate the likelihood. The Laplace approximation is one of the least demanding methods for computing integrals of the form $\int \exp(h(x))dx$. We need only find the maximum of *h* and the second derivative of *h*(*x*) at that point. For the integral in the GLMM likelihood, this can provide a surprisingly good approximation despite the integrand being evaluated at just one point. The maximization step is already familiar from simpler problems.

We can do better with more function evaluations. For these kinds of integrals, Gauss-Hermite quadrature is appropriate. The method approximates integrals of the form $\int h(x) \exp(-x^2)dx$ by $\sum_k w_k f(x_k)$ where the best choice weights w_k and knot-points x_k have been determined. This method is more accurate than the Laplace approach but the computational cost can become prohibitive, particularly for higher dimensional random effects. The Laplace method can be viewed as equivalent to the Gauss-Hermite method with just a single knotpoint.

Experience suggests that numerical integration methods are superior to PQL. The drawback is that they may be time-consuming or impossible to compute for more complex models. The advantage is that we have an approximation to the true likelihood rather than a quasi-likelihood. This means we have more scope and greater confidence in the inference derived from these approaches. They are not perfect since similar issues as with LMMs remain but better than PQL. See McCulloch and Searle (2002) for more discussion.

Bayes: As with LMMs, there is good reason to consider Bayesian methods as

an alternative to the standard likelihood-based methods. There are several advantages. Complex models can be fit with a high degree of accuracy. We can incorporate useful prior information and we have the flexibility to modify the models to allow for nonstandard features. The disadvantages are that these models may require more programming to implement and may take substantial computing resources. Furthermore, one must address technical concerns about the quality of the fit. Finally, the inferential conclusions are of a different form. This is either an advantage or disadvantage depending on your point of view. See Chapter 12 for an introduction to Bayes methods for LMMs. Extending these ideas to GLMMs is not difficult.

We now apply these methods to two examples. The first has a Bernoulli response and the second a Poisson response.

13.3 Binary Response

An experiment was conducted to study the effects of surface and vision on balance. The balance of subjects was observed for two different surfaces and for restricted and unrestricted vision. Balance was assessed qualitatively on an ordinal four-point scale based on observation by the experimenter. Forty subjects were studied, 20 males and 20 females ranging in age from 18 to 38, with heights given in cm and weights in kg. The subjects were tested while standing on foam or a normal surface and with their eyes closed or open or with a dome placed over their head. Each subject was tested twice in each of the surface and eye combinations for a total of 12 measures per subject. The data comes from Steele (1998) via the Australasian Data and Story Library (OzDASL).

For the purposes of this analysis, we will reduce the response to a two-point scale: whether the subject was judged completely stable (=1) or not (=0). We start by defining this response:

```
data(ctsib, package="faraway")
ctsib$stable <- ifelse(ctsib$CTSIB==1,1,0)
```

We can investigate the effects of the treatment variables on stability descriptively. Here is the mean response for the combined conditions:

```
xtabs(stable ~ Surface + Vision, ctsib)/80
```

```
      Vision
Surface closed dome open
foam 0.0000 0.0000 0.1250
norm 0.2125 0.2750 0.8125
```

We have divided by 80 because `xtabs` sums the values for each combination and there are 40 subjects with each combination replicated twice. We see that the normal surface with open vision leads to the highest stability. We can group the data by subject and average over the 12 observations (6 conditions, replicated twice). The plots are seen in Figure 13.1.

```
library(dplyr)
subsum <- ctsib %>% group_by(Subject) %>% summarise(Height=Height[1],
  ↳ Weight=Weight[1], stable=mean(stable), Age=Age[1], Sex=Sex[1])
library(ggplot2)
ggplot(subsum, aes(x=Height,y=stable))+geom_point()
ggplot(subsum, aes(x=Weight,y=stable))+geom_point()
ggplot(subsum, aes(x=Age,y=stable))+geom_point()
```

```
ggplot(subsum, aes(x=Sex,y=stable))+geom_boxplot()
```

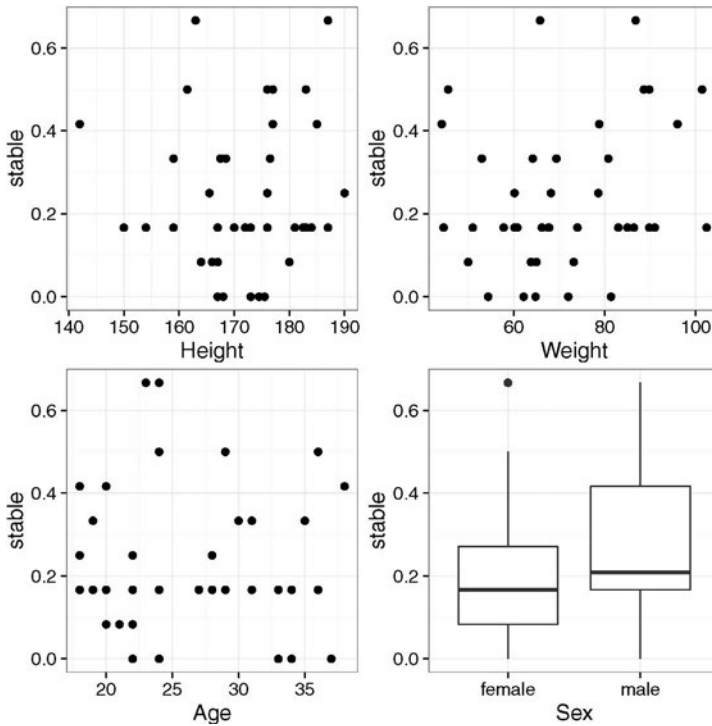


Figure 13.1 *Subject effects for the stability experiment. Response is proportion of stable over treatment conditions.*

We could fit a binomial GLM ignoring the subject information entirely:

```
gf <- glm(stable ~ Sex+Age+Height+Weight+Surface+Vision,binomial,data=
  ↪ ctsib)
summary(gf)
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	7.27745	3.80399	1.91	0.05573
Sexmale	1.40158	0.51623	2.72	0.00663
Age	0.00252	0.02431	0.10	0.91739
Height	-0.09641	0.02684	-3.59	0.00033
Weight	0.04350	0.01800	2.42	0.01567
Surfacenorm	3.96752	0.44718	8.87	< 2e-16
Visiondome	0.36375	0.38322	0.95	0.34252
Visionopen	3.18750	0.41600	7.66	1.8e-14

n = 480 p = 8

Deviance = 295.203 Null Deviance = 526.254 (Difference = 231.051)

This assumes we have 480 independent observations but, in reality, we have only 40 subjects whose responses will be correlated. This analysis is likely to underestimate the standard errors and so exaggerate the significance of the experimental effects. We could also try including a fixed subject factor:

```
gfs <- glm(stable ~ Sex + Age + Height + Weight + Surface + Vision +
  ↪ factor(Subject), binomial,data=ctsib)
```

However, when we examine the summary for this model, we see problems with identifiability. This is because the subject factors cannot be completely distinguished from the four subject-specific measures: sex, age, height and weight. Even if we could get around this problem, it would hardly be appropriate to treat the subject factor as a fixed effect. We do not care about the individual subjects but we are interested in how the four subject measures might affect stability. The experimental subjects are intended as a random sample from the target population. We'd like to know something about the inherent variability in that population that is not explained by measurable variables but we don't care about the specific individuals.

There are a variety of ways of fitting GLMMs in R. First we demonstrate the PQL method implemented in the MASS package:

```
library(MASS)
modpql <- glmmPQL(stable ~ Sex + Age + Height + Weight + Surface +
  ↪ Vision, random=~1|Subject, family=binomial,data=ctsib)
summary(modpql)
```

Random effects:

```
Formula: ~1 | Subject
(Intercept) Residual
```

```
StdDev:      3.0607  0.59062
```

Variance function:

```
Structure: fixed weights
Formula: ~invwt
```

Fixed effects: stable ~ Sex + Age + Height + Weight + Surface + Vision

	Value	Std.Error	DF	t-value	p-value
(Intercept)	15.5715	13.4983	437	1.1536	0.2493
Sexmale	3.3553	1.7526	35	1.9145	0.0638
Age	-0.0066	0.0820	35	-0.0810	0.9359
Height	-0.1908	0.0920	35	-2.0736	0.0455
Weight	0.0695	0.0629	35	1.1052	0.2766
Surfacenorm	7.7241	0.5736	437	13.4665	0.0000
Visiondome	0.7265	0.3259	437	2.2289	0.0263
Visionopen	6.4853	0.5440	437	11.9219	0.0000

The SD for the subject effect is 3.06. We can use the same ideas from logistic regression to interpret this value. We have $\exp(3.06) = 21.3$ so the odds of stability are multiplied by this factor. Hence we can see that there is substantial variation in the inherent stability of individuals. Indeed, this variation is of comparable magnitude to the treatment effects. The residual SD is an artefact of the fitting process and does not exist in the statement of the model.

We see strongly significant surface and vision effects while some other effects have marginally significant p -values. However, this inference is based on the linearized model and rather dubious assumptions as explained in Section 10.2, so these results cannot be relied upon. Furthermore, the Bernoulli response may lead to biased estimates of regression coefficients. Hence, it would be unwise to rely entirely on this analysis without investigating alternative methods of estimation.

The numerical integration-based methods are implemented in the lme4 package. The default choice of method is the Laplace approximation.

```
library(lme4)
```

```
modlap <- glmer(stable ~ Sex + Age + Height + Weight + Surface +
  ↪ Vision + (1|Subject), family=binomial, data=ctsib)
```

Since the Laplace method is a special case of the Gauss-Hermite approximation which can only be more accurate, it is best to attempt this approach. Here we can use the maximum allowable number of quadrature points which is 25:

```
modgh <- glmer(stable ~ Sex + Age + Height + Weight + Surface +
  ↪ Vision + (1|Subject), nAGQ=25, family=binomial, data=ctsib)
```

We have a particularly simple random effects structure so we can easily afford to be profligate in the number of quadrature points (which is certainly more than we need). In more complex examples, we may need to specify much smaller numbers to allow computation in a reasonable time. Start small and increase until the estimates stop changing very much or the computation becomes infeasibly long. Now look at the output:

```
summary(modgh)
```

AIC	BIC	logLik	deviance	df.resid
247.9	285.5	-115.0	229.9	471

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.884	-0.139	-0.020	-0.001	4.902

Random effects:

Groups	Name	Variance	Std.Dev.
Subject	(Intercept)	7.19	2.68

Number of obs: 480, groups: Subject, 40

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	16.17166	12.72107	1.27	0.204
Sexmale	3.09679	1.69612	1.83	0.068
Age	-0.00668	0.07646	-0.09	0.930
Height	-0.19226	0.08895	-2.16	0.031
Weight	0.07515	0.05910	1.27	0.204
Surfacenorm	7.28541	1.05516	6.90	5.0e-12
Visiondome	0.67591	0.52737	1.28	0.200
Visionopen	6.08896	0.97241	6.26	3.8e-10

Notice that we have AIC/BIC values for model comparison purposes. These are not available from PQL because it is not a true likelihood method. As it happens, the parameter estimates are quite similar to PQL which provides some reassurance.

We might ask whether any of the subject-specific variables have an effect. We can test this by fitting a model without these terms and comparing the two:

```
modgh2 <- glmer(stable ~ Surface + Vision + (1|Subject), nAGQ=25,
  ↪ family=binomial, data=ctsib)
```

```
anova(modgh, modgh2)
```

	Df	AIC	BIC	logLik	deviance	Chisq	Chi	Df	Pr(>Chisq)
modgh2	5	247	268	-119	237				
modgh	9	248	286	-115	230	7.37		4	0.12

This uses the standard likelihood-based methods to construct a chi-squared test. We have the same reasons as with LMMs to view these results with some scepticism. Even so, this is a balanced experiment of a reasonable size so this provides some confidence in the result. We see that a simplification to just the treatment variables as fixed effects seems reasonable. If we feel uncomfortable with this conclusion, we

may further point to the minimization of AIC (or BIC) as a justification for choosing the smaller model.

As with all such models, it is wise to check some diagnostics. These can be extracted using `residuals()` and `fitted()` functions. An alternative convenience is:

```
dd <- fortify(modgh2)
```

which extracts the residuals and fitted values and places them in a common data frame with the other variables. This makes the construction of some plots more convenient. For example, we might look at the QQ plots subsetting by the treatment variables:

```
ggplot(dd, aes(sample=.resid))+stat_qq() + facet_grid(Surface~Vision)
```

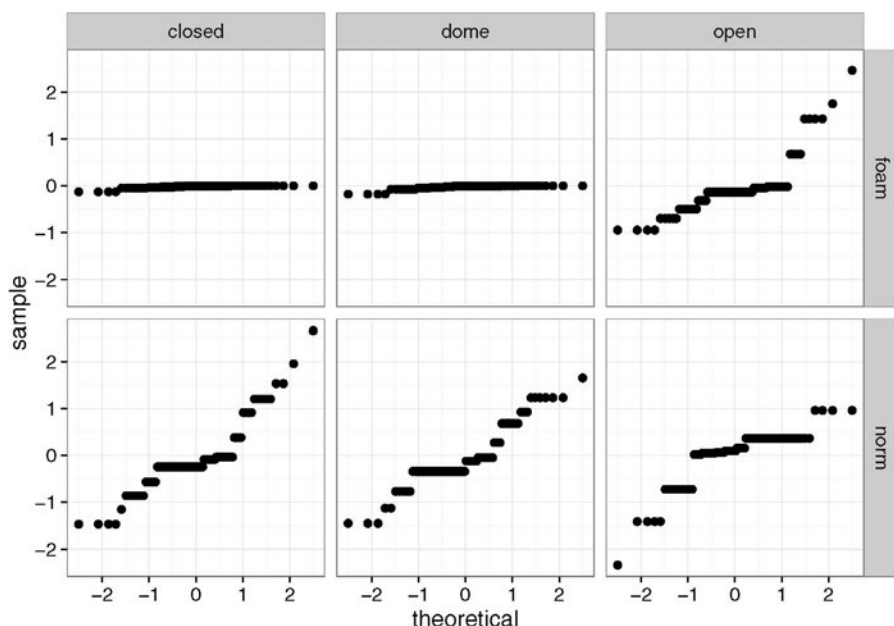


Figure 13.2 *QQ plots subsetting by treatment variables.*

In Figure 13.2, we see that the residuals are close to zero for two of the six combinations. This is because these were universally unstable conditions and have been predicted as such by the model. In the most stable, normal and open condition, larger positive residuals are not seen because there is no headroom for such cases. It would be a mistake to view this plot as indicating heteroscedascity as we have seen there are more convincing explanations for the differences in spread.

We can use INLA for a Bayesian approach to fitting these models. See Section 12.2 for an introduction. For ease of exposition, we use only the surface and vision as fixed effect predictors. The default, noninformative priors, are satisfactory:

```
library(INLA)
formula <- stable ~ Surface + Vision + f(Subject, model="iid")
result <- inla(formula, family="binomial", data=ctsib)
```

We compute the SD for the subject random effect:

```
sigmaalpha <- inla.tmarginal(function(x) 1/sqrt(x), result$marginals.
  ↪ hyperpar$"Precision for Subject")
```

The posterior density for this SD is shown in the first panel of Figure 13.3:

```
x <- seq(0,7,length.out = 100)
sdf <- data.frame(yield = x, density=inla.dmarginal(x, sigmaalpha))
ggplot(sdf, aes(x=yield,y=density))+geom_line()
```

We see that the subject effect is clear since the distribution is well away from zero but there is some uncertainty regarding the size of the effect.

We can produce a numerical summary of the posteriors:

```
restab <- sapply(result$marginals.fixed, function(x) inla.zmarginal(x,
  ↪ silent=TRUE))
restab <- cbind(restab, inla.zmarginal(sigmaalpha,silent=TRUE))
colnames(restab) = c("mu", "norm", "dome", "open", "alpha")
data.frame(restab)
```

	mu	norm	dome	open	alpha
mean	-10.298	7.3641	0.66618	6.1279	3.0248
sd	1.3507	0.92526	0.49873	0.8498	0.62416
quant0.025	-13.172	5.7182	-0.29877	4.6184	1.9838
quant0.25	-11.181	6.6971	0.32503	5.517	2.5756
quant0.5	-10.21	7.3038	0.65785	6.0704	2.9585
quant0.75	-9.334	7.9657	0.99581	6.6771	3.4007
quant0.975	-7.9392	9.3029	1.6579	7.9167	4.429

We could compute similar statistics on the subject random effects but there are too many to display them all. We see that the posterior means are quite similar to the last glmer-based fit. We can plot the posterior densities of the fixed effects as seen in the second panel of Figure 13.3:

```
x <- seq(-2,11,length.out = 100)
rden <- sapply(result$marginals.fixed,function(y) inla.dmarginal(x, y)
  ↪ )[, -1]
ddf <- data.frame(yield=rep(x,3), density=as.vector(rden), treat=gl
  ↪ (3,100, labels=c("norm", "dome", "open")))
ggplot(ddf, aes(x=yield, y=density, linetype=treat))+geom_line()
```

The norm level of surface and the open level of vision are clearly different from the respective reference levels since the densities are well separated from zero. In contrast, we see there may not be much difference between the dome and closed levels of vision as this density overlaps zero. We can compute a “Bayesian p -value” as:

```
2*inla.pmarginal(0,result$marginals.fixed$Visiondome)
[1] 0.17982
```

We have multiplied by two to account for the usual two-sided testing argument. In this context, p -values do not have the same meaning. Nonetheless, it does serve as a measure of how the posterior density relates to zero. This confirms our impression that there is not much difference between the levels.

We can also use STAN for a Bayesian analysis as introduced in Section 12.1. Here is the STAN program we need:

```
data {
  int<lower=0> Nobs;
  int<lower=0> Nsubs;
  int<lower=0> Npreds;
  int<lower=0,upper=1> y[Nobs];
  int<lower=1,upper=Nsubs> subject[Nobs];
```

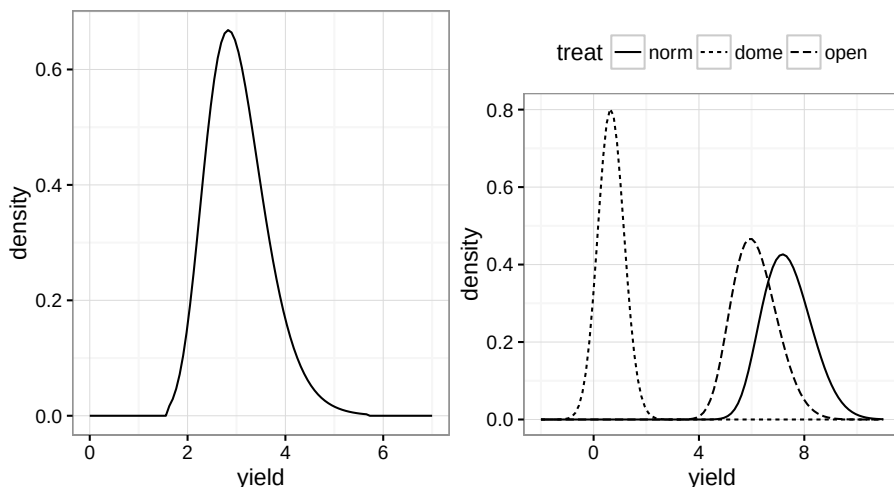


Figure 13.3 Posterior density for the subject SD on the left and posterior densities for the treatment effects on the right.

```

matrix[Nobs,Npreds] x;
}
parameters {
  vector[Nsubs] subeff;
  real<lower=0> sigmasubj;
  vector[Npreds] beta;
}
model {
  subeff ~ normal(0,sigmasubj);
  sigmasubj ~ cauchy(0, 1);
  for(n in 1:Nobs) {
    y[n] ~ bernoulli_logit(x[n]*beta + subeff[subject[n]] );
  }
}

```

We have written this in a generic form so that you could use this for any grouped-by-subject data with a binary response. We use a half-Cauchy prior for the subject SD. This is somewhat more informative but seems justifiable in the context of this data. It also has the advantage of being more transparent.

We need to prepare the data in a format compatible with the data block in the code above. We form the model matrix of fixed effects, X , in advance:

```

xm <- model.matrix(~ Sex + Age + Height + Weight + Surface + Vision,
  ↪ ctsib)
stabledat <- with(ctsib, list(Nobs=nrow(ctsib),
  Nsubs=length(unique(ctsib$Subject)), Npreds=ncol(xm),
  y=stable, subject=Subject, x=xm))

```

We can now run the STAN model. We have broken the process into three steps. The first step translates the STAN code into C++, the second compiles that C++ code and the third runs the MCMC sampler. The advantage of doing it in three stages is that

one is likely only to do the first two once but the third might need to be repeated if the model or data is changed.

```
library(rstan)
rt <- stanc("glmmbin.stan")
sm <- stan_model(stanc_ret = rt, verbose=FALSE)
fit <- sampling(sm, data=stabledat)
```

This will take several minutes to run depending on the quality of your computer.

First we need to check the diagnostics of the MCMC sampling. We plot the chain for the subject SD as this is the parameter most likely to cause problems:

```
traceplot(fit, pars="sigmasubj", inc_warmup=FALSE)
```

The plot (not shown) is entirely satisfactory. We can display a summary for the parameters of interest:

```
print(fit, pars=c("sigmasubj", "beta"))
```

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
sigmasubj	3.60	0.04	0.81	2.28	3.03	3.48	4.09	5.36	515	1.01
beta[1]	19.25	0.73	17.56	-13.12	7.66	18.52	30.11	55.08	585	1.01
beta[2]	3.83	0.10	2.24	-0.45	2.35	3.76	5.27	8.33	531	1.01
beta[3]	-0.01	0.00	0.11	-0.23	-0.08	-0.01	0.06	0.20	836	1.00
beta[4]	-0.23	0.00	0.12	-0.48	-0.30	-0.22	-0.15	-0.01	586	1.01
beta[5]	0.09	0.00	0.08	-0.06	0.03	0.08	0.14	0.24	860	1.00
beta[6]	8.56	0.06	1.33	6.28	7.58	8.43	9.39	11.33	560	1.01
beta[7]	0.75	0.01	0.56	-0.34	0.38	0.74	1.14	1.87	4000	1.00
beta[8]	7.24	0.05	1.24	5.17	6.34	7.13	8.01	9.92	563	1.01

The effective sample sizes are more than satisfactory.

Now we examine the posterior distributions. We extract the parameters of interest and restore the variable names for convenience. The `reshape2` package helps us arrange the data in a format for convenient plotting. We show the estimated densities in Figure 13.4 along with a vertical line at zero.

```
ipars <- data.frame(extract(fit, pars=c("sigmasubj", "beta")))
colnames(ipars)[-1] <- colnames(xm)
library(reshape2)
rdf <- melt(ipars)
ggplot(rdf, aes(x=value))+geom_density() + facet_wrap(~ variable,
  ↪ scales="free")+geom_vline(xintercept=0)
```

We might also be interested in how the subjects in the experiment compare. We extract the subject random effects and sort the posterior means:

```
ppars <- data.frame(extract(fit, pars="subeff"))
sort(colMeans(ppars))
```

subeff.3	subeff.38	subeff.37	subeff.14	
-6.704126	-4.926872	-4.563769	-4.036449	...
...edited...				
subeff.17	subeff.29	subeff.25	subeff.27	
3.488328	5.906336	6.735636	6.924570	

We see that subject 3 is the least stable and subject 27 is the most stable. Since we have access to the posterior distributions, we can readily investigate which difference might be notable.

13.4 Count Response

In this example, we have data from a clinical trial of 59 epileptics. For a baseline, patients were observed for 8 weeks and the number of seizures recorded. The patients

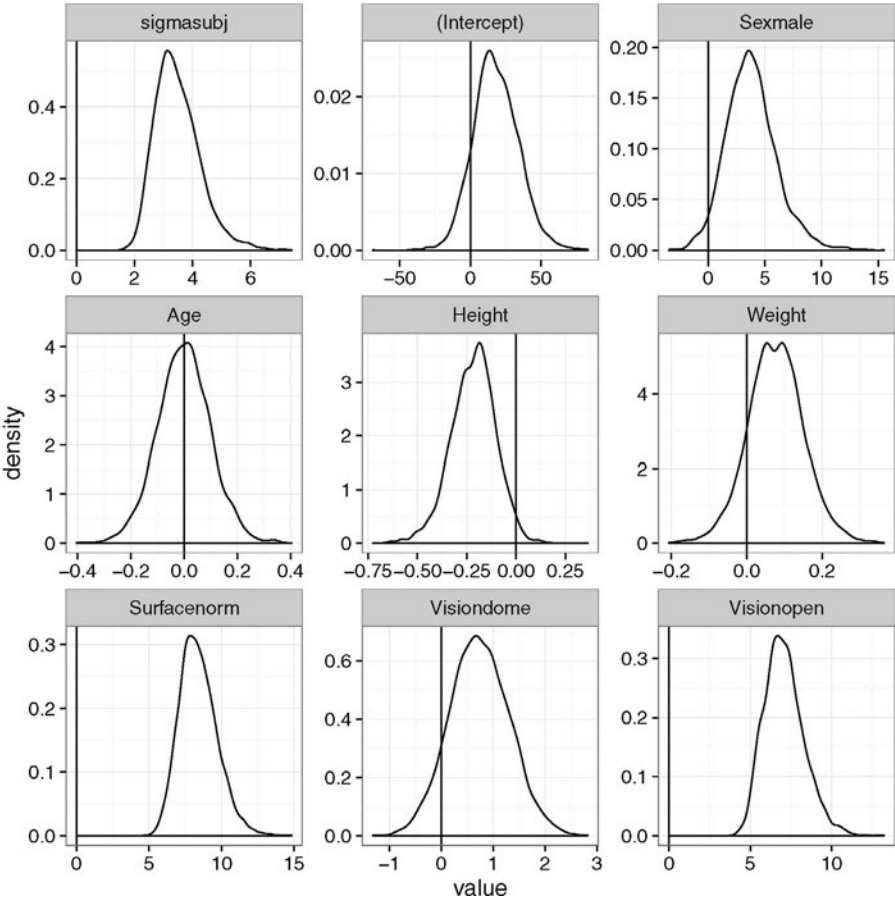


Figure 13.4 Posterior distributions as produced by the STAN fit to the epilepsy data.

were then randomized to treatment by the drug Progabide (31 patients) or to the placebo group (28 patients). They were observed for four 2-week periods and the number of seizures recorded. The data have been analyzed by many authors including Thall and Vail (1990), Breslow and Clayton (1993) and Diggle et al. (2013). Does Progabide reduce the rate of seizures?

First we create some derived variables and then look at the first two patients:

```
data(epilepsy, package="faraway")
epilepsy$period <- rep(0:4, 59)
epilepsy$drug <- factor(c("placebo","treatment")[epilepsy$treat+1])
epilepsy$phase <- factor(c("baseline","experiment")[epilepsy$expind
  ↪ +1])
epilepsy[epilepsy$id < 2.5,]
  seizures id treat expind timeadj age period  drug  phase
1         11  1    0       8   31    0 placebo baseline
2          5  1    0       2   31    1 placebo experiment
```

3	3	1	0	1	2	31	2 placebo experiment
4	3	1	0	1	2	31	3 placebo experiment
5	3	1	0	1	2	31	4 placebo experiment
6	11	2	0	0	8	30	0 placebo baseline
7	3	2	0	1	2	30	1 placebo experiment
8	5	2	0	1	2	30	2 placebo experiment
9	3	2	0	1	2	30	3 placebo experiment
10	3	2	0	1	2	30	4 placebo experiment

Both were not treated ($\text{treat}=0$). The `expind` indicates the baseline phase by 0 and the treatment phase by 1. The length of these time phases is recorded in `timeadj`. We have created three new convenience variables: `period`, denoting the 2- or 8-week periods, `drug` recording the type of treatment in nonnumeric form and `phase` indicating the phase of the experiment.

We now compute the mean number of seizures *per week* broken down by the treatment and baseline vs. experimental period. The `dplyr` package is useful for these types of group summaries:

```
library(dplyr)
epilepsy %>%
  group_by(drug, phase) %>%
  summarise(rate=mean(seizures/timeadj)) %>%
  xtabs(formula=rate ~ phase + drug)
```

	drug	
phase	placebo	treatment
baseline	3.8482	3.9556
experiment	4.3036	3.9839

We see that the rate of seizures in the treatment group actually increases during the period in which the drug was taken. The rate of seizures also increases even more in the placebo group. Perhaps some other factor is causing the rate of seizures to increase during the treatment period and the drug is actually having a beneficial effect. Now we make some plots to show the difference between the treatment and the control. The first plot shows the difference between the two groups during the experimental period only:

```
ggplot(epilepsy, aes(x=period, y=seizures, linetype=drug, group=id))
  ↪ + geom_line() + xlim(1,4) + scale_y_sqrt(breaks=(0:10)^2) +
  ↪ theme(legend.position = "top", legend.direction = "horizontal")
```

We compare the two groups in the left panel of Figure 13.5 and find little to choose between them. The square-root transform is used to stabilize the variance; this is often used with count data. Now we compare the average seizure rate to the baseline for the two groups:

```
ratesum <- epilepsy %>%
  group_by(id, phase, drug) %>%
  summarise(rate=mean(seizures/timeadj))
library(tidyr)
comsum <- spread(ratesum, phase, rate)
ggplot(comsum, aes(x=baseline, y=experiment, shape=drug)) + geom_point
  ↪ () + scale_x_sqrt() + scale_y_sqrt() + geom_abline(intercept=0,
  ↪ slope=1) + theme(legend.position = "top", legend.direction = "
  ↪ horizontal")
```

A treatment effect, if one exists, is not readily apparent. Now we fit GLMM models. Patient #49 is unusual because of the high rate of seizures observed. We exclude it:

```
epilo <- filter(epilepsy, id != 49)
```

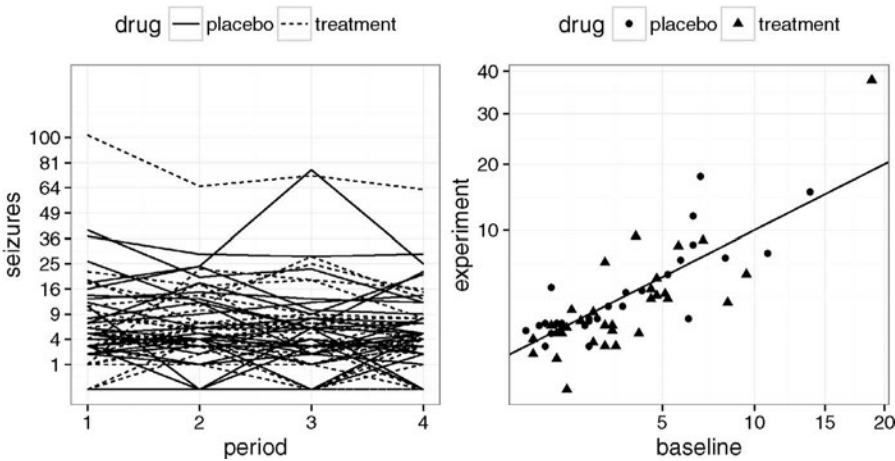


Figure 13.5 Seizures per 2-week period on a square-root scale with treatment group shown as solid lines and the placebo group shown as dotted lines in the plot on the left. Mean seizures per week is shown on the right. We compare the baseline period with the experimental period, distinguishing those who receive treatment or control.

Excluding a case should not be taken lightly. It is worth repeating the analysis with and without this subject. For projects where the analyst works with producers of the data, it will be possible to discuss substantive reasons for excluding cases. Exclusion of cases should always be reported and not concealed.

It is worth starting with a GLM even though the model is not correct due to the grouping of the observations. We must use an *offset* as explained in Section 5.3 to allow for the difference in lengths in the baseline and treatment periods.

```
modglm <- glm(seizures ~offset(log(timeadj)) + expind + treat + I(
  ↪ expind*treat), family=poisson, data=epilo)
sumary(modglm)
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.3476	0.0341	39.57	< 2e-16
expind	0.1118	0.0469	2.39	0.017
treat	-0.1068	0.0486	-2.20	0.028
I(expind * treat)	-0.3024	0.0697	-4.34	0.000014

n = 290 p = 4
Deviance = 2411.550 Null Deviance = 2485.110 (Difference = 73.560)

The interaction term is the primary parameter of interest. All the subjects were untreated in the baseline, even the ones who were subsequently treated. This means that the main effect for treatment does properly measure the response to treatment because it includes the baseline period. As we have observed already, we suspect the response may have been different during the baseline time and the active period of the experiment. The interaction term represents the effect of the treatment during the baseline period after adjustment. In the output above we see that this interaction seems highly significant and negative (which is good since we want to reduce

seizures). But this inference is suspect because we have made no allowance for the correlated responses within individuals. The p -value is far smaller than it should be.

We might also consider allowing for overdispersion in the response by using a quasi-Poisson model as discussed in Section 9.4. However, this is a different consideration to the correlated response.

We move through the estimation options in the same order as with the binary response example earlier, starting with PQL:

```
library(MASS)
modpql <- glmmPQL(seizures ~ offset(log(timeadj)) + expind + treat + I(
  ↪ expind*treat), random = ~1|id, family=poisson, data=epilo)
summary(modpql)
Formula: ~1 | id
          (Intercept) Residual
StdDev:      0.68197   1.6054

Variance function:
Structure: fixed weights
Formula: ~invwt
Fixed effects: seizures ~ offset(log(timeadj)) + expind + treat + I(expind*treat)
              Value Std.Error   DF t-value p-value
(Intercept)   1.08079  0.143701  230   7.5211  0.0000
expind         0.11184  0.075767  230   1.4761  0.1413
treat         -0.00894  0.200244   56  -0.0446  0.9646
I(expind * treat) -0.30238  0.112689  230  -2.6834  0.0078
```

The parameter estimates are comparable to the GLM but the standard errors are larger as might be expected given that the correlated response has been allowed for. As with the binary response example, we still have some doubts about the accuracy of the inference. This is a particular concern when some count responses are small. A further concern is the problematic meaning of the residual SD as such a term does not appear in the statement of the model. Also we lack an AIC due to the quasi-ness of the likelihood. Even so, we do see a significant negative interaction effect indicating that the drug is effective.

Numerical quadrature can also be used. We use Gauss-Hermite in preference to Laplace as the model random effect structure is simple and so the computation is fast even though we have used the most expensive $nAGQ=25$ setting.

```
library(lme4)
modgh <- glmer(seizures ~ offset(log(timeadj)) + expind + treat + I(
  ↪ expind*treat) + (1|id), nAGQ=25, family=poisson, data=epilo)
summary(modgh)
Random effects:
Groups Name      Variance Std.Dev.
id      (Intercept) 0.515    0.718
Number of obs: 290, groups: id, 58
```

```
Fixed effects:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   1.03600    0.14126   7.33  2.2e-13
expind         0.11184    0.04688   2.39   0.017
treat         -0.00815    0.19652  -0.04   0.967
I(expind * treat) -0.30239    0.06971  -4.34  1.4e-05
```

We see that the interaction effect is significant. Notice that the estimate of this effect

has been quite consistent over all the estimation methods so we draw some confidence from this. We have

```
exp(-0.302)
```

```
[1] 0.73934
```

So the drug is estimated to reduce the rate of seizures by about 26%. However, the subject SD is more than twice the drug effect of -0.3 at 0.718 . This indicates that the expected improvement in the drug is substantially less than the variation between individuals. Interpretation of the main effect terms is problematic in the presence of an interaction. For example, the treatment effect reported here represents the predicted difference in the response during the baseline period (i.e., `expind=0`). Since none of the subjects are treated during the baseline period, we are reassured to see that this effect is not significant. However, this does illustrate the danger in naively presuming that this is the treatment effect.

We can also take a Bayesian approach, starting with STAN. We can use almost the same code as the binary response example except we need to add

```
vector[Nobs] offset;
```

in the data block and replace the model line with

```
y[n] ~ poisson_log(log(offset[n])+x[n]*beta + subeff[subject[n]] );
```

We prepare the data into the required format using:

```
epilo$id[epilo$id == 59] <- 49
```

```
xm <- model.matrix( ~ expind + treat + I(expind*treat), epilo)
epildat <- with(epilo, list(Nobs=nrow(epilo), Nsubs=length(unique(id)),
                           Npreds=ncol(xm),
                           y=seizures,
                           subject=id,
                           x=xm, offset=timeadj))
```

We've renumbered case 59 into the previously deleted case 49 slot. This is ugly but we need the subjects to be consecutively numbered.

Assuming that the code is placed in a file called `glmmpois.stan`, we translate, compile and run the sampler:

```
library(rstan)
```

```
rt <- stanc("glmmpois.stan")
```

```
sm <- stan_model(stanc_ret = rt, verbose=FALSE)
```

```
fit <- sampling(sm, data=epildat)
```

We can check the sampling properties of the chain by

```
traceplot(fit, pars="sigmasubj", inc_warmup=FALSE)
```

We've made the plot only for subject SD since this is the one most likely to cause problems. In this case, the plot (not shown) is satisfactory. We can review the posterior distributions:

```
ipars <- data.frame(rstan::extract(fit, pars=c("sigmasubj", "beta")))
colnames(ipars) <- c("subject", "intercept", "expind", "treat", "
  ↪ interaction")
```

We plot the two most interesting posterior distributions as seen in Figure 13.6:

```
ggplot(ipars, aes(x=subject))+geom_density()
ggplot(ipars, aes(x=interaction))+geom_density() +geom_vline(
  ↪ xintercept=0)
```

We can see the subject SD is very clearly different from zero and that it is centered on about 0.7 . The interaction effect (or drug effect) is negative centered on about -0.3 . This looks clearly less than zero.

We can construct a convenient summary of the results including a sort of p -value.

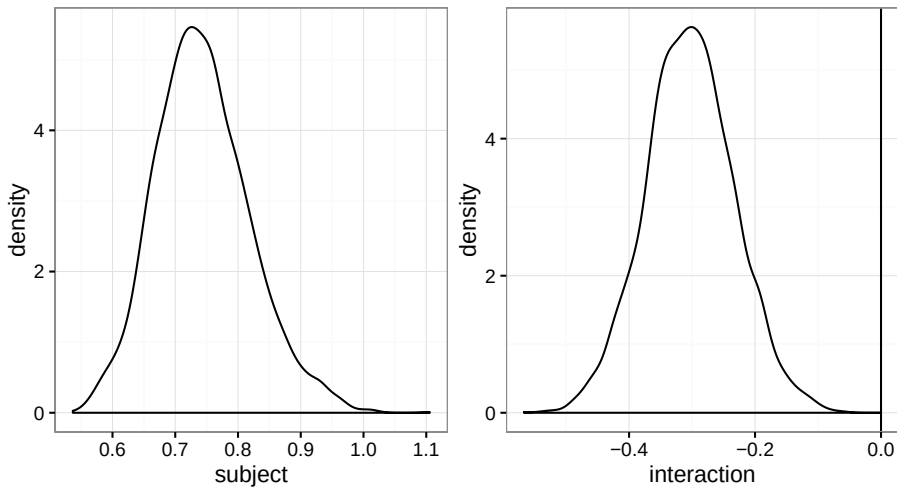


Figure 13.6 Posterior distribution of the subject SD and drug effect for the epilepsy data.

```
bayespval <- function(x) {p <- mean(x > 0); 2*min(p,1-p)}
smat <- apply(ipars, 2, function(x) c(mean(x), quantile(x,c(0.025,
  ↪ 0.975))), bayespval(x))
row.names(smat) <- c("mean", "LCB", "UCB", "pvalue")
t(smat)
```

	mean	LCB	UCB	pvalue
subject	0.743504	0.613355	0.90935	0.0000
intercept	1.021433	0.736583	1.30687	0.0000
expind	0.110912	0.017704	0.20310	0.0210
treat	0.005876	-0.386557	0.39178	0.9695
interaction	-0.301699	-0.434090	-0.17111	0.0000

We see that the posterior means reported here are quite similar to the estimates computed earlier using likelihood methods. The Bayesian analysis seems rather more confident that there is a drug effect since this posterior distribution is well separated from zero.

The same model can also be fit by INLA in a straightforward way:

```
formula <- seizures ~offset(log(timeadj)) + expind + treat + I(expind*
  ↪ treat) + f(id,model="iid")
result <- inla(formula, family="poisson", data = epil0)
```

We obtain a summary of the posteriors as:

```
sigmaalpha <- inla.tmarginal(function(x) 1/sqrt(x), result$marginals.
  ↪ hyperpar$"Precision for id")
restab <- sapply(result$marginals.fixed, function(x) inla.zmarginal(x,
  ↪ silent=TRUE))
restab <- cbind(restab, inla.zmarginal(sigmaalpha,silent=TRUE))
colnames(restab) = c("mu", "expind", "treat", "interaction", "alpha")
data.frame(restab)
```

	mu	expind	treat	interaction	alpha
mean	1.036	0.11178	-0.0081489	-0.30246	0.72583
sd	0.14196	0.046891	0.19753	0.06973	0.070948

```

quant0.025 0.75556 0.019681 -0.39732 -0.4396 0.59944
quant0.25 0.94023 0.079891 -0.14164 -0.34985 0.67548
quant0.5 1.0354 0.11151 -0.0093067 -0.3028 0.72079
quant0.75 1.1304 0.14314 0.12302 -0.25579 0.77093
quant0.975 1.3133 0.20349 0.3787 -0.16626 0.87796

```

We see that the results are similar to those obtained previously. We observe that the 95% credible interval for the interaction is $(-0.44, -0.17)$ so we are sure that this parameter differs from zero as in the STAN analysis. We can compute further numerical and graphical summaries as in previous examples obtaining very similar results.

13.5 Generalized Estimating Equations

The advantage of the quasi-likelihood approach as described in Section 9.4 compared to GLMs was that we did not need to specify the distribution of the response. We only needed to give the link function and the variance. We can adapt this approach for repeated measures and/or longitudinal studies. Let Y_i be a vector of random variables representing the responses on a given individual or cluster and let $EY_i = \mu_i$ which is then linked to the linear predictor using $g(\mu_i) = x_i^T \beta$, where g is a link function appropriate to the response type and x_i is the predictor vector.

As with the quasi-likelihood, we also need to specify a variance function $a(\cdot)$:

$$\text{var } Y_i = \phi a(\mu_i)$$

Certain choices of $a(\cdot)$ will be sensible depending on the type of response. The ϕ is a scale parameter which may be set to one if not needed.

In addition, we must also specify how the responses within an individual or cluster are correlated with each other. We set a *working correlation matrix* $R_i(\alpha)$ depending on a parameter α which we will estimate. This results in a *working covariance matrix* for Y_i :

$$V_i = \phi A_i^{1/2} R_i(\alpha) A_i^{1/2}$$

where A_i is a diagonal matrix formed from $a(\mu_i)$.

Given estimates of ϕ and α , we can estimate β by setting the (multivariate) score function to zero and solving:

$$\sum_i \left(\frac{\partial \mu_i}{\partial \beta} \right)^T V_i^{-1} (Y_i - \mu_i) = 0$$

These equations can be regarded as the multivariate analogue of those used for the quasi-likelihood models described in Section 9.4. Since $\text{var } Y$ also depends on α , we substitute any consistent estimate of α in this equation and still obtain an estimate as asymptotically efficient as if α were known. A similar set of equations can be derived representing the score with respect to α , which may be similarly solved. We iterate between estimating α and β until we converge at a solution.

These are called *generalized estimating equations* (GEE). Note that no specification of the distribution has been necessary which makes the fitting and specification much simpler. The estimates of β are consistent even if the variance is misspecified.

We use the `geepack` package as described in Højsgaard et al. (2005). The `gee` package can also fit these models with somewhat different features.

We reanalyze the stability dataset:

```
data(ctsib, package="faraway")
ctsib$stable <- ifelse(ctsib$CTSIB==1,1,0)
library(geepack)
modgeep <- geeglm(stable ~ Sex + Age + Height + Weight + Surface +
  ↪ Vision, id=Subject, corstr="exchangeable", scale.fix=TRUE,
  ↪ data=ctsib, family=binomial)
```

We have specified the same fixed effects as in the corresponding GLMM earlier. The grouping variable is specified by the `id` argument. Only simple groups are allowed while nested grouping variables cannot be accommodated easily in this function. We must choose the correlation structure within each group. If we choose no correlation, then the problem reduces to a standard GLM. Several choices are available. For this data, it seems reasonable to assume that any pair of observations from the same subject have the same correlation. This is known as an *exchangeable* correlation or, equivalently, *compound symmetry*. We have chosen to fix the scale parameter at the default value of 1 to ensure maximum compatibility with the GLMM fit. Otherwise, there would not be a strong reason to fix this. Let us now examine the output:

summary(modgeep)

```
Coefficients:
              Estimate Std.err Wald Pr(>|W|)
(Intercept)   8.6233   5.9199   2.12   0.145
Sexmale       1.6449   0.9035   3.31   0.069
Age          -0.0121   0.0480   0.06   0.802
Height       -0.1021   0.0424   5.80   0.016
Weight        0.0437   0.0340   1.65   0.199
Surfacenorm   3.9163   0.5668  47.74  4.9e-12
Visiondome    0.3589   0.4040   0.79   0.374
Visionopen    3.1799   0.4606  47.66  5.1e-12
```

Scale is fixed.

Correlation: Structure = exchangeable Link = identity

Estimated Correlation Parameters:

```
              Estimate Std.err
alpha        0.218   0.0447
Number of clusters: 40 Maximum cluster size: 12
```

We can see that the estimated correlation between observations on the same subject is 0.22. The standard error of 0.04 indicates that we can be quite sure there is a correlation in the responses within individuals.

The standard errors are constructed using a *sandwich estimator* as described in Section 8.5. These are typically, but not always, larger than the naive standard errors from the likelihood calculations. These standard errors can be used to construct Wald statistics. We see that the treatment factors, surface and vision, are significant. Height and possibly gender are marginally significant. This part of the conclusion is similar to our GLMM results.

There is one clear difference with the GLMM output: the estimates for the GEE are about half the size of the GLMM β s. This is to be expected. GLMMs model the

data at the subject or individual level. The correlation between the measurements on the individual is generated by the random effect. Thus the β s for the GLMM represent the effect on an individual. A GEE models the data at the population level. The β s for a GEE represent the effect of the predictors averaged across all individuals with the same predictor values. GEEs do not use random effects but model the correlation at the marginal or correlation level.

The testing for vision is not entirely satisfactory since it has three levels meaning two tests—one being highly significant and the other not at all. If we want a single test for the significance of vision, we need to refit the model without vision and make the standard anova-type comparison:

```
modgeep2 <- geeglm(stable ~ Sex + Age + Height + Weight + Surface, id
  ↪ =Subject, corstr="exchangeable", scale.fix=TRUE, data=ctsib,
  ↪ family=binomial)
anova(modgeep2, modgeep)
```

Analysis of 'Wald statistic' Table

```
Model 1 stable ~ Sex + Age + Height + Weight + Surface + Vision
Model 2 stable ~ Sex + Age + Height + Weight + Surface
  Df    X2 P(>|Chi|)
1  2 58.4  2.1e-13
```

As expected, we see that vision is strongly significant.

The `geepack` package also offers the possibility of modeling an ordinal response with clusters using the `ordgee()` function. This would be appropriate for the original form of this data where the response is measured on a four-point scale.

We can also model the epilepsy data:

```
data(epilepsy, package="faraway")
```

We exclude the 49th case as before. An autoregressive AR(1) model for the correlation structure is most natural since consecutive measurements will be more correlated than measurements separated in time. Note that this does require that the clusters be sorted in time order — they are in this case.

```
modgeep <- geeglm(seizures ~offset(log(timeadj)) + expind + treat + I(
  ↪ expind*treat), id=id, family=poisson, corstr="ar1", data=
  ↪ epilepsy, subset=(id!=49))
summary(modgeep)
```

Coefficients:

	Estimate	Std.err	Wald	Pr(> W)
(Intercept)	1.3138	0.1616	66.10	4.4e-16
expind	0.1509	0.1108	1.86	0.173
treat	-0.0797	0.1983	0.16	0.688
I(expind * treat)	-0.3987	0.1745	5.22	0.022

Estimated Scale Parameters:

	Estimate	Std.err
(Intercept)	10.6	2.35

Correlation: Structure = ar1 Link = identity

Estimated Correlation Parameters:

	Estimate	Std.err
alpha	0.783	0.0519

Number of clusters: 58 Maximum cluster size: 5

The drug effects, as measured by the interaction term, has a just significant effect.

The dispersion parameter is estimated as 10.6. This means that if we did not account for the overdispersion, the standard errors would be much larger. The AR(1) correlation structure can be seen in the working correlation where adjacent measurements have 0.78 correlation.

Further analysis would involve an investigation of alternative correlation structures, the age covariate and any trend during the experimental period. The analysis of this dataset is discussed in Diggle et al. (2013).

Further Reading: McCulloch and Searle (2002) have some coverage of GLMMs as well as more material on GLMs. Hardin and Hilbe (2003) give a book-length treatment of GEEs. Diggle et al. (2013) discuss both topics.

Exercises

1. The `ohio` data concern 536 children from Steubenville, Ohio and were taken as part of a study on the effects of air pollution. Children were in the study for 4 years from ages 7 to 10. The response was whether they wheezed or not. The variables are:

resp an indicator of wheeze status (1 = yes, 0 = no)

id an identifier for the child

age 7 yrs = -2, 8 yrs = -1, 9 yrs = 0, 10 yrs = 1

smoke an indicator of maternal smoking at the first year of the study (1 = smoker, 0 = nonsmoker)

- (a) Do any of the mothers in the study change their smoking status during the period of observation?
- (b) Construct a table that shows proportion of children who wheeze for 0, 1, 2, 3 or 4 years broken down by maternal smoking status.
- (c) Make plot which shows how the proportion of children wheezing changes by age with a separate line for smoking and nonsmoking mothers.
- (d) Group the data by child to count the total (out of four) years of wheezing. Fit a binomial GLM to this response to check for a maternal smoking effect. Does this prove there is a smoking effect or could there be another plausible explanation?
- (e) Fit a model for each individual response using a GLMM fit using penalized quasi-likelihood. Describe the effects of age and maternal smoking. How do the odds of wheezing change numerically over time?
- (f) Now fit the same model but using adaptive Gaussian-Hermit quadrature. Compare to the previous model fit.
- (g) Use `INLA` to fit the same model. What does this model say about the effect of age and maternal smoking?
- (h) Use `STAN` to fit the same model. Check the MCMC diagnostics and again discuss the age and maternal smoking effects.

- (i) Fit the model using GEE. Use an autoregressive rather than exchangeable error structure. Compare the results to the previous model fits. In your model, what indicates that a child who already wheezes is likely to continue to wheeze?
 - (j) What is your overall conclusion regarding the effect of age and maternal smoking? Can we trust the GLM result or are the GLMM models preferable?
2. The National Youth Survey collected a sample of 11–17 year olds, 117 boys and 120 girls, asking questions about marijuana usage. The data is presented in `potuse`.
- (a) Plot the total number of people falling into each usage category as it varies over time separately for each sex.
 - (b) Condense the levels of the response into whether the person did or did not use marijuana that year. Turn the year into a numerical variable. Fit a GLMM for the now binary response with an interaction between sex and year as a predictor using Gauss-Hermite quadrature. Comment on the effect of sex.
 - (c) Fit a reduced model without sex and use it to test for the significance of sex in the larger model.
 - (d) Fit a model with year as a factor. Should this model be preferred to the model with year as just a linear term? Interpret the estimated effects in the year as a factor version of the model.
 - (e) Fit GEE version of the model and compare it to the analogous GLMM fit.
3. Components are attached to an electronic circuit card assembly by a wave-soldering process. The soldering process involves baking and preheating the circuit card and then passing it through a solder wave by conveyor. Defects arise during the process. The design is 2^{7-3} with three replicates and the data is found in `wavesolder`.
- (a) Plot the data to show how the number of defects varies with the predictors.
 - (b) Fit a Poisson GLM to the individual runs with the number of defects as the response and main effects for all the predictors. How can you tell that this model is inadequate? Fit a comparable quasi-Poisson GLM. What difference does this make to the significance of the predictors?
 - (c) Sum the defects within each replicate group of three and fit a quasi-Poisson GLM to these sums. Compare the fitted model to the previous one.
 - (d) Fit a GEE model to the individual defect responses with a fixed scale that allows for an autoregressive correlation structure within the groups (assuming that the replicates are in time order). Is it reasonable to fix the scale?
 - (e) Now refit without a fixed scale. Is there any evidence of a correlation between successive replicates?
 - (f) Finally fit a GEE model with an independent correlation structure within the replicates. Compare this model to the quasi-Poisson GLM fit.
4. The `nitrofen` data in `boot` package come from an experiment to measure the reproductive toxicity of the pesticide nitrofen on a species of zooplankton called *Ceriodaphnia dubia*. Each animal produced three broods in which the number of

live offspring was recorded. Fifty animals in total were used and divided into five batches. Each batch was treated in a solution with a different concentration of the pesticide.

- (a) Plot the total number of live offspring as they vary with concentration and comment. Now plot the numbers for each brood, taking care to distinguish the different broods. Is the trend within each brood the same?
 - (b) Fit a GLMM for the number of live offspring within each brood that varies with concentration and brood number (including an interaction). The model should take account of the dependence between observations from the same animal. Describe what the model says about how the number of live offspring change with concentration for the different broods.
 - (c) Fit an equivalent GEE model and compare it to the GLMM result.
5. The `toenail` data comes from a multicenter study comparing two oral treatments for toenail infection. Patients were evaluated for the degree of separation of the nail. Patients were randomized into two treatments and were followed over seven visits: four in the first year and yearly thereafter. The patients have not been treated prior to the first visit so this should be regarded as the baseline.
- (a) Calculate the proportion of patients with a normal or severe condition broken down by treatment and visit number. Plot these proportions and comment.
 - (b) Fit a GLMM for the outcome as a function of an interaction between the visit and the treatment. Since the two groups are selected at random, there should be no difference at the first visit. Does this model show a significant difference at this baseline (first visit)?
 - (c) Test for a significant treatment effect by fitting a model without treatment and comparing to the previous model.